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Small Sounding Rocket Apogee Roll Rate

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Nomenclature

Mnemonic	Definition
C_L	Roll moment coefficient $=\frac{2L}{\rho S d V^2}$,
C_{Lp}	Roll moment coefficient damping derivative near $M=0$,
$C_{L\delta}$	Roll moment coefficient driving derivative near $M=0$,
d	Aerodynamic reference length,
g	Acceleration due to gravity,
h	Altitude,
I_R	Roll Moment of inertia,
L	Rolling moment,
M	Mach number,
p	Roll rate,
S	Aerodynamic reference area,
t	Time,
V	Flight speed,
ρ	Atmospheric mass density,
δ	Fin cant angle,
$()_A$	Value of $()$ at apogee, and
$()_B$	Value of $()$ at burnout.

Introduction

In recent years university programs teaching engineering by having students design, build and fly small sounding rockets have proliferated. Mostly these university rockets have apogees less than 10 km. For these rockets, there are two major reasons why apogee roll rate is important. First, many experiments flown aboard small university sounding rockets with apogees in the troposphere are sensitive to roll rate, especially near apogee where payloads linger longest. For example, consider a sensor that images the scenes around it. The frame rate needed will depend on roll rate. Second, many sounding rockets deploy their drogue parachutes near apogee at low dynamic pressure. Roll rate can be a consideration when there is a risk of fouling riser lines.

Estimation of an approximate roll rate at lower altitudes near burnout is easy: Neglect the roll moment of inertia, balance the damping and driving torques, and solve for the helical roll rate. But, as dynamic pressure falls during ascent, the roll moment of

inertia acquires more and more importance until it dominates the solution. The roll rate near apogee could, of course, be found directly by numerically integrating the roll equation of motion, but this can be time-consuming. In this note we seek an approximate result amenable to rapid evaluation.

Assumptions

A few simplifying assumptions are needed for this analysis.

1. Take the trajectory to be vertical. Even though this cannot be not true near apogee, there is no mechanism for trajectory curvature there to change roll rate. Apogee is assumed to lie in the troposphere.
2. The rocket vehicle is slender with a very small roll moment of inertia.
3. Assume the angle of attack is small while the dynamic pressure remains large enough to engender significant aerodynamic roll moments. This implies negligible aerodynamic nonlinearities.
4. Aerodynamic drag is much smaller than weight near apogee. Low dynamic pressure near apogee ensures this will be so.
5. All fin panels have the same cant angle.
6. Only roll rate, flight speed and altitude vary with time during ascent. While other parameters such as atmospheric density also vary, this variation has negligible influence on roll rate during most of ascent because it is not part of the classical helix solution. Density and the stability derivatives do matter near apogee. Their local values at apogee are used in the analysis.

Analysis

Consistent with the Assumptions, the roll equation of motion is per Ref's. (1 and2)

$$I_R \frac{dp}{dt} = \frac{1}{2}\rho_A S d V^2 [C_{Lp}(\frac{pd}{2V}) + C_{L\delta}\delta]. \qquad (1)$$

Now, let

$$G = -\frac{\rho_A S d^2 C_{Lp}}{4 g I_R} \quad \text{and} \quad H = \frac{\rho_A S d C_{L\delta}\delta}{2 g I_R}. \qquad (2)$$

Note that the burnout roll rate can be estimated by neglecting the roll moment of inertia:

$$p_B \cong \frac{H}{G} V_B. \qquad (3)$$

Then,

$$\frac{dp}{dt} + gGVp = gHV^2. \qquad (4)$$

Next, change the independent variable from time to velocity. Then, near apogee, since gravity is the primary force changing the velocity, $\frac{dV}{dt} \cong - g$ giving

$$\frac{dp}{dV} - GVp = - HV^2 . \tag{5}$$

An integrating factor J for Eq. (5) can be found to be:

$$J = \exp(- \frac{GV^2}{2}). \tag{6}$$

After integrating Eq. (5) and substituting from Eq. (3), we find that

$$\frac{p_A}{p_B} = \exp(\frac{- GV_B^2}{2}) + \frac{G}{V_B} \int_0^{V_B} V^2 \exp(\frac{- GV^2}{2}) dV . \tag{7}$$

Unfortunately, the integral in Eq. (7) cannot be written in closed form except in the limit as GV_B goes to ∞. But, Eq. (7) shows that the dimensionless independent variable for this problem is GV_B^2, and that the dimensionless dependent variable is $\frac{p_A}{p_B}$. Numerical integration can then be used to generate Figure 1 below which is applicable to any small university sounding rocket.

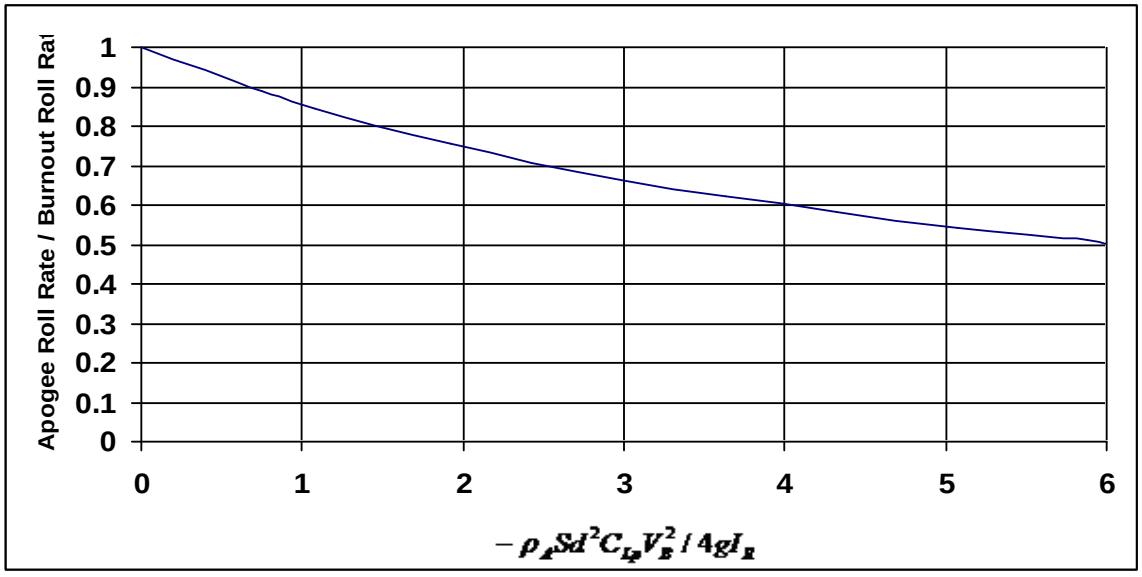


Figure1 Apogee Roll Rate

Conclusion

Figure 1 can be used to estimate the typical university sounding rocket's apogee roll rate. The reader is cautioned that this approximation is only valid for small rockets having apogees well within the tropopause.

References

1. Babister, A. W., *Aircraft Dynamic Stability and Response*, Pergamon Press Ltd, Oxford, 1980.
2. Etkin, B., *Dynamics of Flight, Stability and Control*, John Wiley and Sons, New York, 1959.